BdMO 2026 National Camp F.Eqns PSet

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§1 In-Lecture Problems

Problem 1.1. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that:

$$f\left(x^2+y\right)=f\left(x^{27}+2y\right)+f\left(x^4\right)$$

for all $x,y\in\mathbb{R}$.

Problem 1.2. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that:

$$f(x+yf(x))+x=f(xf(y))+f(y+f(x))$$

for all $x,y\in\mathbb{R}$.

Problem 1.3. Find all functions $f:\mathbb{R}\to\mathbb{R}$ such that:

$$f(xf(y)+x)=xy+f(x).$$

Problem 1.4. Solve over $\mathbb{Z}$ the functional equation

$$f(2a)+2f(b)=f(f(a+b)).$$

Problem 1.5. Find all functions $f:\mathbb{R}\to\mathbb{R}$ satisfying

$$f(x^2+y)=f(f(x)-y)+4f(x)y$$

for all real numbers x and y.

Problem 1.6. Find all functions $f:\mathbb{R}\to\mathbb{R}$ for which

$$f(xf(x)+f(y))=f(x)^2+y$$

holds for all real numbers x and y.

§2 Buffer Problems

Problem 2.1. Solve over $\mathbb{R}$ the functional equation

$$f\left(x^2-y\right)+2yf(x)=f(f(x))+f(y).$$

Problem 2.2. Find all strictly increasing functions $f:\mathbb{Z}_{\geq 0}\to\mathbb{Z}_{>0}$ satisfying $f(2)=7$ and

$$f(mn)=f(m)f(n)+f(m)+f(n)$$

for all non-negative integers m and n.

Problem 2.3. Find all $f : \mathbb{R}^+ \to \mathbb{R}^+$ such that for all **distinct** positive real numbers x, y, z, we have

$$f(x)^2 - f(y)f(z) = f\left(x^y\right) f(y)f(z) \left[f\left(y^z\right) - f\left(z^x\right)\right].$$

Problem 2.4. Find all functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f\left(xf(x+y)\right) = f\left(yf(x)\right) + x^2.$$

§3 Resources

- Introduction to Functional Equations
- Monster Functional Equations
- 100 Functional Equations - Amir Hossein Parvadi
- Functional Equations - Pang-Cheng Wu